

Lid-Driven Flow with Explicit Projection Method

ME 7751 Homework 5

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Scripts Included

`lid_driven_flow.py`, `finite_differences.py`, `iterative_solve.py`, `plotting.py`, `plot_csv.py`, `part2a.sh`, `part2b.sh`, `part3a.sh`, `part3b.sh`

Usage Instructions

All Bash scripts call functionality to `lid_driven_flow.py` and `plot_csv.py`. `lid_driven_flow.py` imports functions from `finite_differences.py`, `iterative_solve.py`, and `plotting.py`, which include reused code from previous assignments. Run any script to regenerate all the images used for its corresponding part. Run `python lid_driven_flow.py -h` to see available command-line arguments.

Problem Statement

Consider 2-D flow of viscous, incompressible Newtonian fluid with density ρ and dynamic viscosity μ , driven by a moving lid with horizontal velocity U . The channel has height and length H , making the domain $(x, y) \in [0, H] \times [0, H]$. The flow is governed by the momentum conservation equations.

$$\rho \left(\frac{\partial u}{\partial t} + \frac{\partial uu}{\partial x} + \frac{\partial uv}{\partial y} \right) = \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} - \frac{\partial p}{\partial x} \quad (1)$$

$$\rho \left(\frac{\partial v}{\partial t} + \frac{\partial vu}{\partial x} + \frac{\partial vv}{\partial y} \right) = \frac{\partial \tau_{yx}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} - \frac{\partial p}{\partial y} \quad (2)$$

For an incompressible fluid,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \quad (3)$$

For a Newtonian fluid, the viscous stress tensor τ is linearly related to the strain rate by:

$$\tau_{xx} = \mu \left(2 \frac{\partial u}{\partial x} \right) \quad (4)$$

$$\tau_{yy} = \mu \left(2 \frac{\partial v}{\partial y} \right) \quad (5)$$

$$\tau_{xy} = \tau_{yx} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \quad (6)$$

No-slip boundary conditions are applied at the walls ($u = 0, v = 0, \frac{\partial p}{\partial n} = 0$) except the top wall ($y = H$) where $u = U, v = 0, \frac{\partial p}{\partial n} = 0$. By setting the length scale $H = 1$, the time scale $H/U = 1$ (or $U = 1$), and the mass scale $\rho H^3 = 1$ (or $\rho = 1$), we effectively non-dimensionalized all given quantities. Additionally, the Reynold number $Re = \rho U H / \mu = U H / \nu$ is completely determined by the kinematic viscosity $\nu = \mu / \rho$. From now on, we will simplify future equations by setting $\rho = 1$, meaning $\mu = \nu, Re = \nu^{-1}$.

1a. Description of Explicit Projection Method

The explicit projection method is a numerical method used to solve the incompressible Navier-Stokes equations. The main idea behind this method is to split the solution process into two steps: an explicit step to compute an intermediate velocity field, and a projection step to enforce the incompressibility constraint. To implement the explicit projection method, we first discretize the momentum equations using finite differences on a rectangular staggered mesh. The time derivative is discretized using explicit Euler time-marching.

The explicit step involves solving the momentum equations without considering the pressure term, resulting in an intermediate velocity field that may not satisfy the incompressibility condition. Using finite differences and the explicit Euler method, we can express the intermediate velocity fields u^*, v^* as:

$$u^* = u^n + \Delta t \left(-\frac{\delta u u}{\delta x} - \frac{\delta u v}{\delta y} + \frac{\delta \tau_{xx}}{\delta x} + \frac{\delta \tau_{xy}}{\delta y} \right)^n \quad (7)$$

$$v^* = v^n + \Delta t \left(-\frac{\delta v u}{\delta x} - \frac{\delta v v}{\delta y} + \frac{\delta \tau_{yx}}{\delta x} + \frac{\delta \tau_{yy}}{\delta y} \right)^n \quad (8)$$

Here, $\delta/\delta x, \delta/\delta y$ represent finite difference operators which will be defined later on. The projection step then involves solving a Poisson equation for the pressure, which is derived from the incompressibility constraint. Using the intermediate velocity fields, we can express the pressure Poisson equation as:

$$\nabla^2 p^{n+1} = \frac{1}{\Delta t} \left(\frac{\delta u^*}{\delta x} + \frac{\delta v^*}{\delta y} \right) \quad (9)$$

Since solving the Poisson equation is computationally expensive, we use successive overrelaxation to find the pressure field (discussed in a previous homework assignment). Once the pressure is obtained, we can correct the intermediate velocity fields to enforce incompressibility:

$$u^{n+1} = u^* - \Delta t \frac{\delta p^{n+1}}{\delta x} \quad (10)$$

$$v^{n+1} = v^* - \Delta t \frac{\delta p^{n+1}}{\delta y} \quad (11)$$

1b. Description of Staggered Mesh and Finite Differences

To discretize the governing equations, we use a staggered mesh where the velocity components u and v and the pressure p are all defined at different locations as shown in **Figure 1**, reducing numerical instabilities and ensures that the incompressibility constraint is satisfied. For a square domain of height H , we divide it into a grid of $N \times N$ cells with grid spacing $h = H/N$. A layer of ghost cells is added around the domain to apply boundary conditions which are discussed later. Thus, the grid points for the velocity and pressure fields are defined as follows:

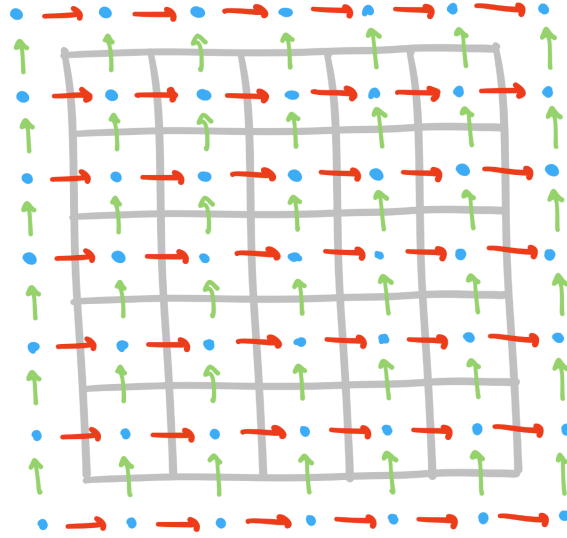


Figure 1: Staggered mesh arrangement for velocity and pressure. The u velocity components are defined at the centers of the vertical cell faces (red), the v velocity components are defined at the centers of the horizontal cell faces (green), and the pressure is defined at the cell centers (blue). Cell faces are shown in gray. A layer of ghost cells is added around the domain to apply boundary conditions.

$$u_{i,j} \text{ at } (x_i, y_j) = \left(ih, \left(j - \frac{1}{2} \right) h \right), \quad i = 0, 1, \dots, N, \quad j = 0, 1, \dots, N+1 \quad (12)$$

$$v_{i,j} \text{ at } (x_i, y_j) = \left(\left(i - \frac{1}{2} \right) h, jh \right), \quad i = 0, 1, \dots, N+1, \quad j = 0, 1, \dots, N \quad (13)$$

$$p_{i,j} \text{ at } (x_i, y_j) = \left(\left(i - \frac{1}{2} \right) h, \left(j - \frac{1}{2} \right) h \right), \quad i = 0, 1, \dots, N+1, \quad j = 0, 1, \dots, N+1 \quad (14)$$

Using this staggered arrangement, we can define finite difference operators for the spatial derivatives. The intermediate velocity fields $u_{i,j}^*$ are computed at each grid point (i, j) using the following central difference approximations:

$$u_{i,j}^* = u_{i,j}^n + \Delta t \left\{ - \frac{(u_{i+1,j}^n + u_{i,j}^n)^2 - (u_{i-1,j}^n + u_{i,j}^n)^2}{h} \right. \quad (15)$$

$$\left. - \frac{(u_{i,j+1}^n + u_{i,j}^n)(v_{i+1,j}^n + v_{i,j}^n) - (u_{i,j-1}^n + u_{i,j}^n)(v_{i+1,j-1}^n + v_{i,j-1}^n)}{4h} \right. \quad (16)$$

$$\left. + 2\nu \frac{u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{h^2} \right. \quad (17)$$

$$\left. + \nu \left(\frac{v_{i+1,j}^n - v_{i,j}^n}{h^2} + \frac{u_{i,j+1}^n - u_{i,j}^n}{h^2} \right) + \nu \left(\frac{v_{i+1,j-1}^n - v_{i,j-1}^n}{h^2} + \frac{u_{i,j-1}^n - u_{i,j}^n}{h^2} \right) \right\} \quad (18)$$

Here, (15)-(18) correspond to $\delta uu/\delta x$, $\delta uv/\delta y$, $\delta \tau_{xx}/\delta x$, $\delta \tau_{xy}/\delta y$ respectively. Similarly, $v_{i,j}^*$ is computed using the following central difference approximations:

$$v_{i,j}^* = v_{i,j}^n + \Delta t \left\{ - \frac{(v_{i+1,j}^n + v_{i,j}^n)(u_{i,j+1}^n + u_{i,j}^n) - (v_{i-1,j}^n + v_{i,j}^n)(u_{i-1,j+1}^n + u_{i-1,j}^n)}{4h} \right. \quad (19)$$

$$\left. - \frac{(v_{i,j+1}^n + v_{i,j}^n)^2 - (v_{i,j-1}^n + v_{i,j}^n)^2}{h} \right. \quad (20)$$

$$+ 2\nu \frac{v_{i,j+1}^n - 2v_{i,j}^n + v_{i,j-1}^n}{h^2} \quad (21)$$

$$\left. + \nu \left(\frac{u_{i,j+1}^n - u_{i-1,j+1}^n}{h^2} + \frac{v_{i+1,j}^n - v_{i,j}^n}{h^2} \right) + \nu \left(\frac{u_{i,j}^n - u_{i-1,j}^n}{h^2} + \frac{v_{i-1,j}^n - v_{i-2,j}^n}{h^2} \right) \right\} \quad (22)$$

Here, (19)-(22) correspond to $\delta v u / \delta x$, $\delta v v / \delta y$, $\delta \tau_{yy} / \delta y$, $\delta \tau_{yx} / \delta x$ respectively.

1c. Description of Pressure Poisson Equation and SOR Method

The pressure Poisson equation is discretized using the following central difference approximation:

$$\nabla^2 p^{n+1} = \frac{1}{\Delta t} \left(\frac{u_{i,j}^* - u_{i-1,j}^*}{h} + \frac{v_{i,j}^* - v_{i,j-1}^*}{h} \right) \quad (23)$$

Where ∇^2 is now the discretized Laplacian operator. To solve the pressure Poisson equation, we use the successive overrelaxation (SOR) method which iteratively updates the pressure field until convergence.

$$p_{i,j}^{n,(k+1)} = p_{i,j}^{n,(k)} + \frac{\alpha}{4} \left[p_{i-1,j}^{n,(k+1)} + p_{i+1,j}^{n,(k)} + p_{i,j-1}^{n,(k+1)} + p_{i,j+1}^{n,(k)} - h^2 \text{RHS}_{i,j} \right] \quad (24)$$

Here, $\text{RHS}_{i,j}$ is the right-hand side of (23), α is the overrelaxation parameter, and k is the SOR iteration number. In the actual implementation, GitHub Copilot was used to update the pressure field using a "red-black" vectorization of SOR (as opposed to a double for loop). This involves updating the pressure values at the "red" grid points (where $i + j$ is even) and "black" grid points (where $i + j$ is odd) separately, allowing for more efficient memory access and faster convergence.

The pressure Poisson equation enforces the incompressibility constraint by ensuring that the divergence of the velocity field is zero. This is achieved by correcting the intermediate velocity fields using the pressure gradient, which effectively projects the velocity field onto a divergence-free space. These corrected velocity fields at the next time step are given by:

$$u^{n+1} = u^* - \Delta t \frac{p_{i,j}^{n+1} - p_{i-1,j}^{n+1}}{h} \quad (25)$$

$$v^{n+1} = v^* - \Delta t \frac{p_{i,j}^{n+1} - p_{i,j-1}^{n+1}}{h} \quad (26)$$

1d. Description of Boundary Conditions

For the lid-driven flow problem, we apply no-slip boundary conditions at the walls, which means that the velocity of the fluid at the boundary is equal to the velocity of the wall. Specifically, we have:

$$u_{i,0} = -u_{i,1}, \quad v_{i,0} = 0 \quad (\text{bottom wall}) \quad (27)$$

$$u_{i,N+1} = 2U - u_{i,N}, \quad v_{i,N} = 0 \quad (\text{top wall}) \quad (28)$$

$$u_{0,j} = 0, \quad v_{0,j} = -v_{1,j} \quad (\text{left wall}) \quad (29)$$

$$u_{N,j} = 0, \quad v_{N+1,j} = -v_{N,j} \quad (\text{right wall}) \quad (30)$$

While $v_{i,0}, v_{i,N}, u_{0,j}, u_{N,j}$ are set to zero to satisfy the no-slip condition, $u_{i,0}$ and $u_{i,N+1}$ are set using a linear extrapolation to ensure that the velocity at the boundary is equal to the velocity of the wall. The pressure boundary conditions are set such that the normal derivative of pressure at the walls is zero, which can be implemented by setting the pressure values in the ghost cells equal to the adjacent interior cells.

$$p_{i,0} = p_{i,1} \quad (\text{bottom wall}) \quad (31)$$

$$p_{i,N+1} = p_{i,N} \quad (\text{top wall}) \quad (32)$$

$$p_{0,j} = p_{1,j} \quad (\text{left wall}) \quad (33)$$

$$p_{N+1,j} = p_{N,j} \quad (\text{right wall}) \quad (34)$$

2. Steady State Solution for $Re = 100$ and $Re = 400$

To compute the steady state solution for $Re = 100$ and $Re = 400$, we run the solver with $N = 129, \Delta t = 5 \times 10^{-4}, \alpha = 1.5$ until the velocity field converges to a steady state. This is determined by checking if $\max_{i,j}(u_{i,j}^{n+1} - u_{i,j}^n) < \Delta t \cdot 10^{-6}$ and $\max_{i,j}(v_{i,j}^{n+1} - v_{i,j}^n) < \Delta t \cdot 10^{-6}$. Within each time step, the pressure field is iteratively solved for according to "red-black" vectorization of SOR until $\max_{i,j}(p_{i,j}^{n,(k+1)} - p_{i,j}^{n,(k)}) < \Delta t \cdot 10^{-6}$. The resulting steady state streamlines, divergence, and centerline velocities are plotted in **Figure 2**.

As expected, the streamlines show a single primary vortex near the center of the domain and multiple smaller vortices in the corners of the domain, matching what we see in real lid-driven flow. The divergence plots show that the velocity field is approximately divergence-free, with values close to zero throughout the domain. Sharp peaks in divergence are present at the top corners of the domain likely due to numerical errors at the corners. The centerline velocity plots show that the numerical solution matches well with the expected velocity profiles for both $Re = 100$ and $Re = 400$.

3. Qualitative Spatial Convergence Rate

To analyze the spatial convergence rate of the solver, we run the solver for $Re = 100$ and $Re = 400$ with different grid resolutions ($N = 33, 65, 129$) while keeping $\Delta t = 5 \times 10^{-4}$ and $\alpha = 1.5$ fixed. The resulting steady state centerline velocities are plotted in **Figure 3**. Through visual inspection, we can see that the centerline velocities approach the literature values, indicating that the solution is converging to a more accurate representation of the flow field.

4. Quantitative Spatial Convergence Rate

To quantitatively analyze the spatial convergence rate of the solver, we compute the mean absolute error (MAE) between the numerical solution and the literature values for the centerline velocities. Since the literature values are only available at discrete points along the centerline, we use linear interpolation to estimate the numerical solution at those points. The absolute error at each available literature point along the vertical centerline (comparing x-velocities) and horizontal centerline (comparing y-velocities) is computed. Afterward, the mean of these absolute errors is taken to obtain the MAE for each grid resolution. The resulting MAE values for $Re = 100$ and are plotted against the grid spacing h in **Figure 4**.

It is clear to see that the spatial convergence rate doesn't match the theoretical second-order convergence rate of the finite difference scheme. This is potentially due to the fact that Δt is held constant while h is reduced, meaning that the temporal error may be dominating the spatial error and preventing us from observing the expected spatial convergence rate. Additionally, the presence of sharp peaks in divergence at the corners of the domain may also be contributing to numerical errors that affect the convergence rate. To better observe the expected spatial convergence rate, we could try reducing Δt as we refine the grid to ensure that the temporal error is sufficiently small compared to the spatial error.

5. Discussion of Method Stability and Stability Criteria

In order to maintain stability of the explicit projection method, we need to satisfy certain criteria related to the time step Δt , grid spacing h , and Reynolds number Re . The main stability criteria for the explicit projection method are:

$$\Delta t < \frac{h}{U} \quad (\text{CFL condition}) \quad (35)$$

$$\Delta t < \frac{h^2}{4\nu} \quad (\text{diffusion stability condition}) \quad (36)$$

The CFL condition ensures that the fluid does not advect more than one grid cell in a single time step, while the diffusion stability condition ensures that the viscous diffusion does not cause numerical instability. For our simulations, we choose $\Delta t = 5 \times 10^{-4}$ which satisfies both of these conditions for the given grid resolutions and Reynolds numbers. Additionally, the choice of the overrelaxation parameter α in the SOR method can also affect the convergence of the pressure solver. A common choice for α is between 1.0 and 2.0, with $\alpha = 1.5$ often providing good convergence rates for the pressure Poisson equation.

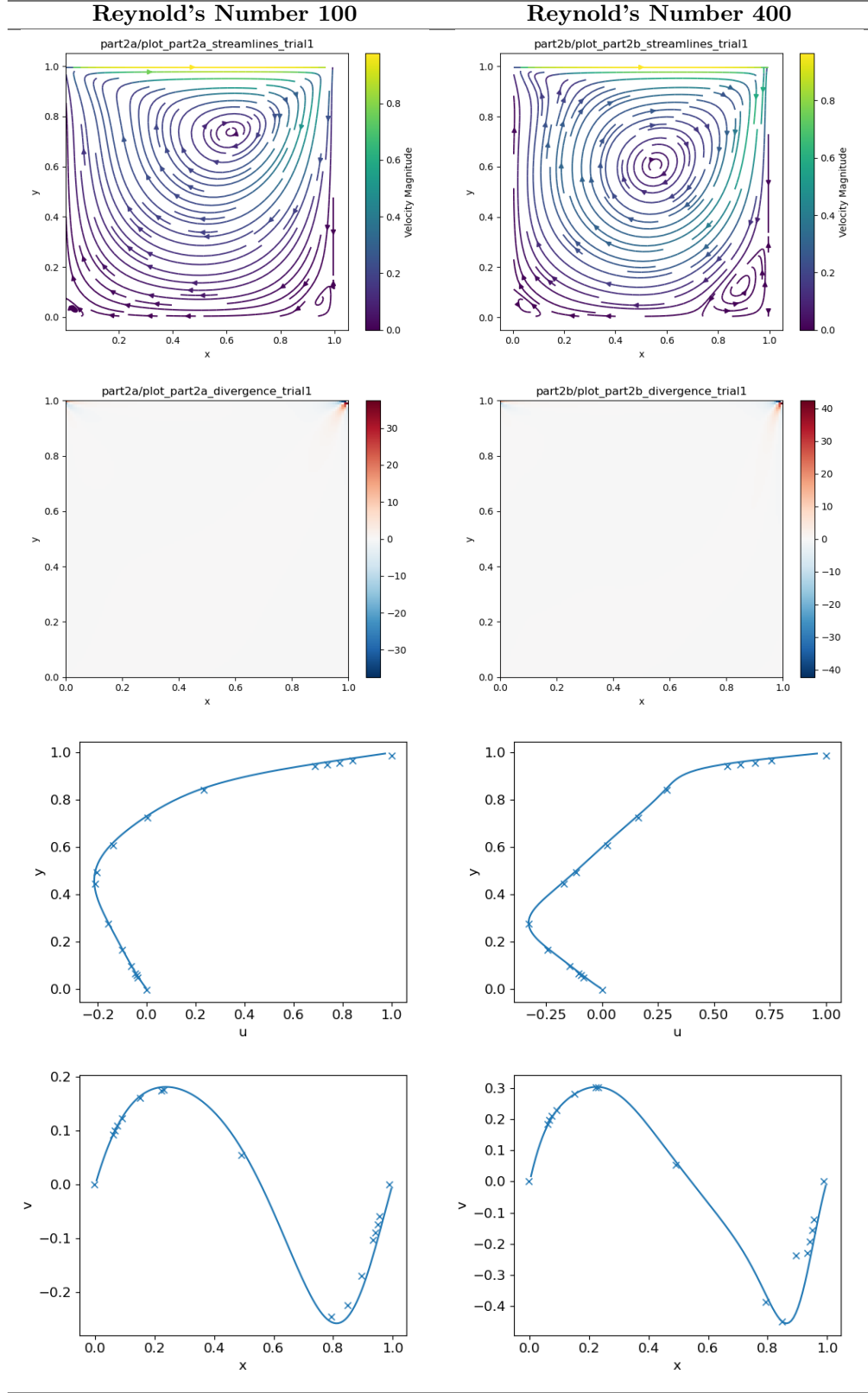


Figure 2: Steady state streamlines (first row), divergence (second row), vertical centerline x-velocities (third row), and horizontal centerline y-velocities (fourth row) for $Re = 100$ and $Re = 400$ with $N = 129, \Delta t = 5 \times 10^{-4}, \alpha = 1.5$. Literature values for the centerline velocities are marked with X's for comparison.

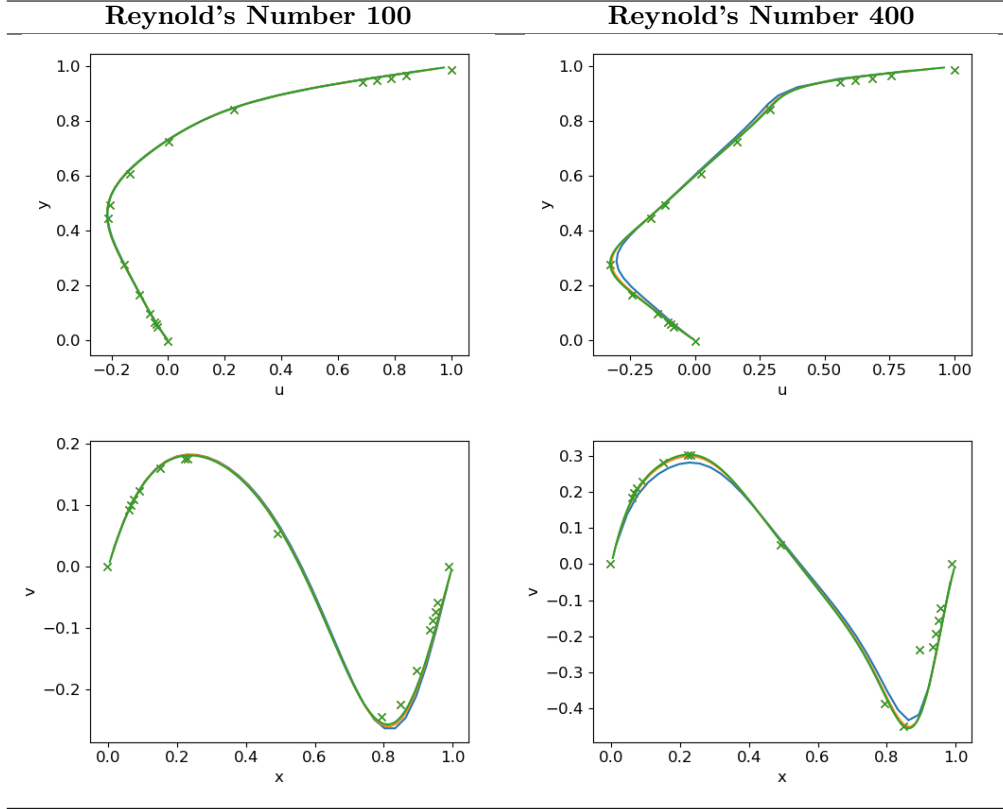


Figure 3: Centerline x-velocity profiles for $Re = 100$ and $Re = 400$ with different grid resolutions ($N = 33$ (blue), 65 (orange), 129 (green)) while keeping $\Delta t = 5 \times 10^{-4}$ and $\alpha = 1.5$ fixed. Literature values for the centerline velocities are marked with X's for comparison.

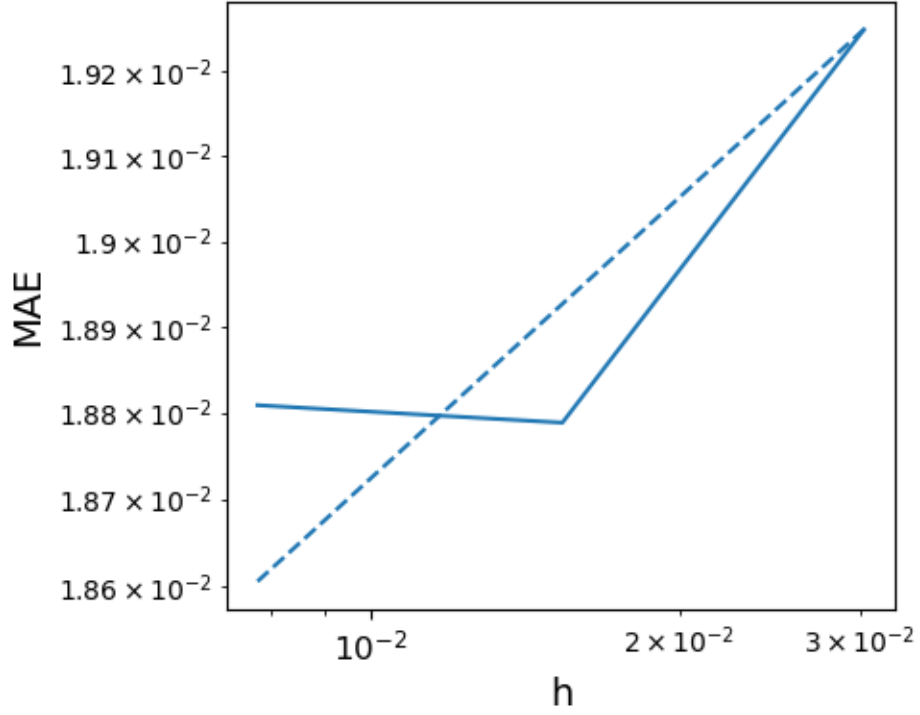


Figure 4: Mean absolute error (MAE) between the numerical solution and literature values for the centerline velocities at $Re = 100$ and $Re = 400$ with different grid resolutions ($N = 33, 65, 129$) while keeping $\Delta t = 5 \times 10^{-4}$ and $\alpha = 1.5$ fixed. The plot is on a log-log scale to better visualize the convergence rate. The slope of the dotted line is 0.025, indicating a convergence rate of approximately 2.5%.